Capstone Checkpoint 5.1 - Agentic TOC Search, Table Lookup and Compute

Oracle, vector, and hybrid experiment comparison. Generated 2026-09-22 19:13 EDT.

19	5	361	19	0
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

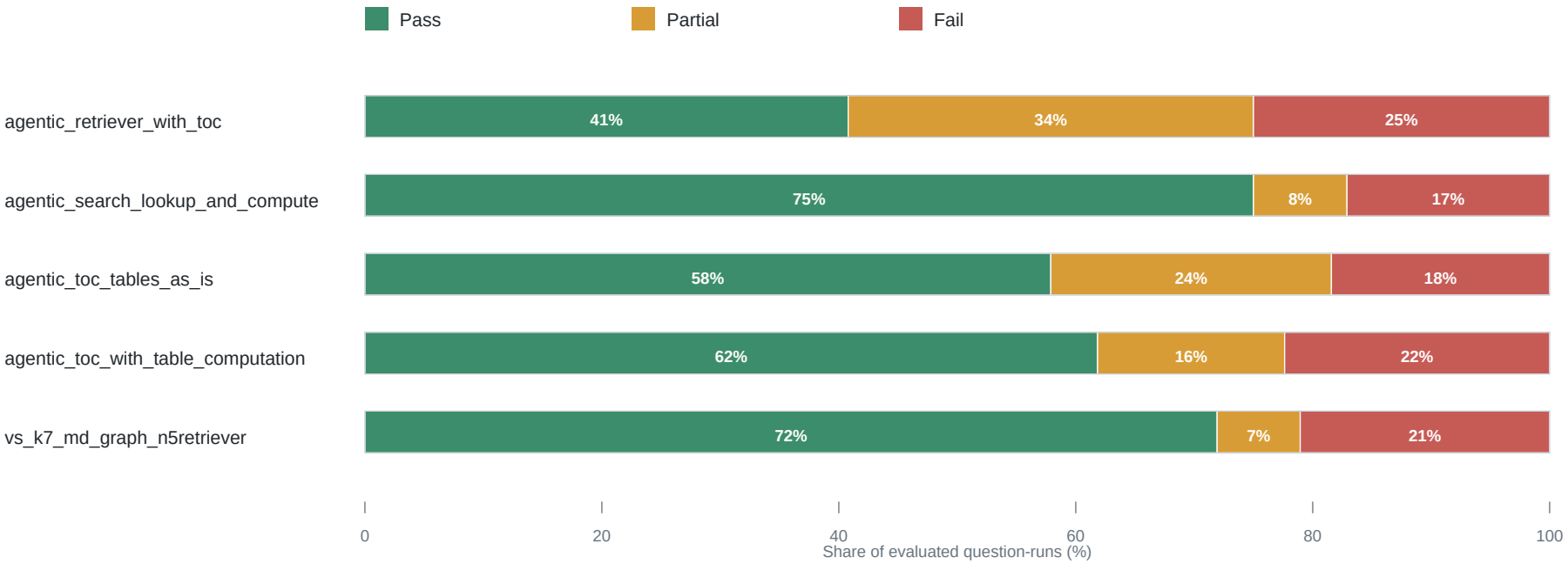
Summary

The highest observed mean quality score was 0.789 for agentic_search_lookup_and_compute across 4 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 100.0% for agentic_search_lookup_and_compute. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 57.5 for Exhaustive list or time filter under agentic_search_lookup_and_compute. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes

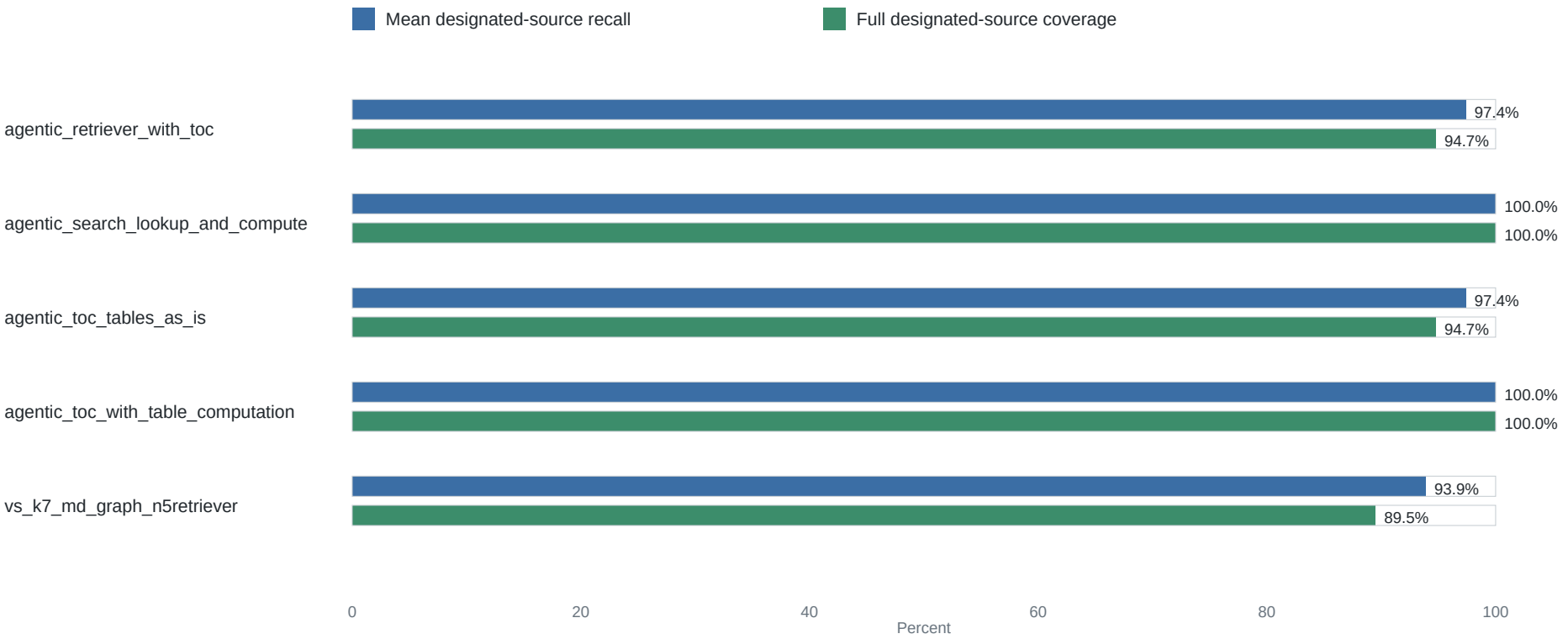


Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
agentic_retriever_with_toc	4	76	40.8%	34.2%	25.0%	0.579	0.032	97.4%	94.7%	13.3k
agentic_search_lookup_and_compute	4	76	75.0%	7.9%	17.1%	0.789	0.026	100.0%	100.0%	44.2k
agentic_toc_tables_as_is	4	76	57.9%	23.7%	18.4%	0.697	0.023	97.4%	94.7%	28.3k
agentic_toc_with_table_computation	4	76	61.8%	15.8%	22.4%	0.697	0.023	100.0%	100.0%	43.5k
vs_k7_md_graph_n5retriever	3	57	71.9%	7.0%	21.1%	0.754	0.045	93.9%	89.5%	260.7k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
agentic_retriever_with_toc	31	0	41	4
agentic_search_lookup_and_compute	57	0	19	0
agentic_toc_tables_as_is	40	4	32	0
agentic_toc_with_table_computation	47	0	29	0
vs_k7_md_graph_n5retriever	38	3	13	3

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

	agentic_retriever_with_toc	agentic_search_lookup_and_c..	agentic_toc_tables_as_is	agentic_toc_with_table_comp..	vs_k7_md_graph_n5retriever
GT01	1.00	1.00	1.00	1.00	1.00
GT02	0.62	1.00	0.75	0.62	1.00
GT03	0.00	0.00	0.12	0.00	0.00
GT04	1.00	1.00	1.00	1.00	1.00
GT05	1.00	1.00	1.00	1.00	1.00
GT06	0.25	1.00	0.12	1.00	1.00
GT07	0.25	0.12	0.50	0.00	0.33
GT08	0.00	1.00	0.62	0.38	0.67
GT09	0.38	1.00	0.75	0.88	1.00
GT10	0.50	1.00	0.50	0.50	0.00
GT11	0.00	0.00	0.12	0.12	0.00
GT12	0.62	1.00	1.00	1.00	1.00
GT13	0.88	1.00	1.00	1.00	1.00
GT14	1.00	1.00	1.00	1.00	1.00
GT15	1.00	1.00	1.00	1.00	1.00
GT16	0.38	0.38	0.38	0.38	0.50
GT17	0.62	1.00	0.50	1.00	1.00
GT18	1.00	1.00	0.88	0.88	1.00
GT19	0.50	0.50	1.00	0.50	0.83

Fail (0) Partial (0.5) Pass (1)

Question	agentic_retriever_with_toc	agentic_search_lookup_and_compa	agentic_toc_tables_as_is	agentic_toc_with_table_computation	s_k7_md_graph_n5retriever
GT01	1.000	1.000	1.000	1.000	1.000
GT02	0.625	1.000	0.750	0.625	1.000
GT03	0.000	0.000	0.125	0.000	0.000
GT04	1.000	1.000	1.000	1.000	1.000
GT05	1.000	1.000	1.000	1.000	1.000
GT06	0.250	1.000	0.125	1.000	1.000
GT07	0.250	0.125	0.500	0.000	0.333
GT08	0.000	1.000	0.625	0.375	0.667
GT09	0.375	1.000	0.750	0.875	1.000
GT10	0.500	1.000	0.500	0.500	0.000
GT11	0.000	0.000	0.125	0.125	0.000
GT12	0.625	1.000	1.000	1.000	1.000
GT13	0.875	1.000	1.000	1.000	1.000
GT14	1.000	1.000	1.000	1.000	1.000
GT15	1.000	1.000	1.000	1.000	1.000
GT16	0.375	0.375	0.375	0.375	0.500
GT17	0.625	1.000	0.500	1.000	1.000
GT18	1.000	1.000	0.875	0.875	1.000
GT19	0.500	0.500	1.000	0.500	0.833

Per-question article recall

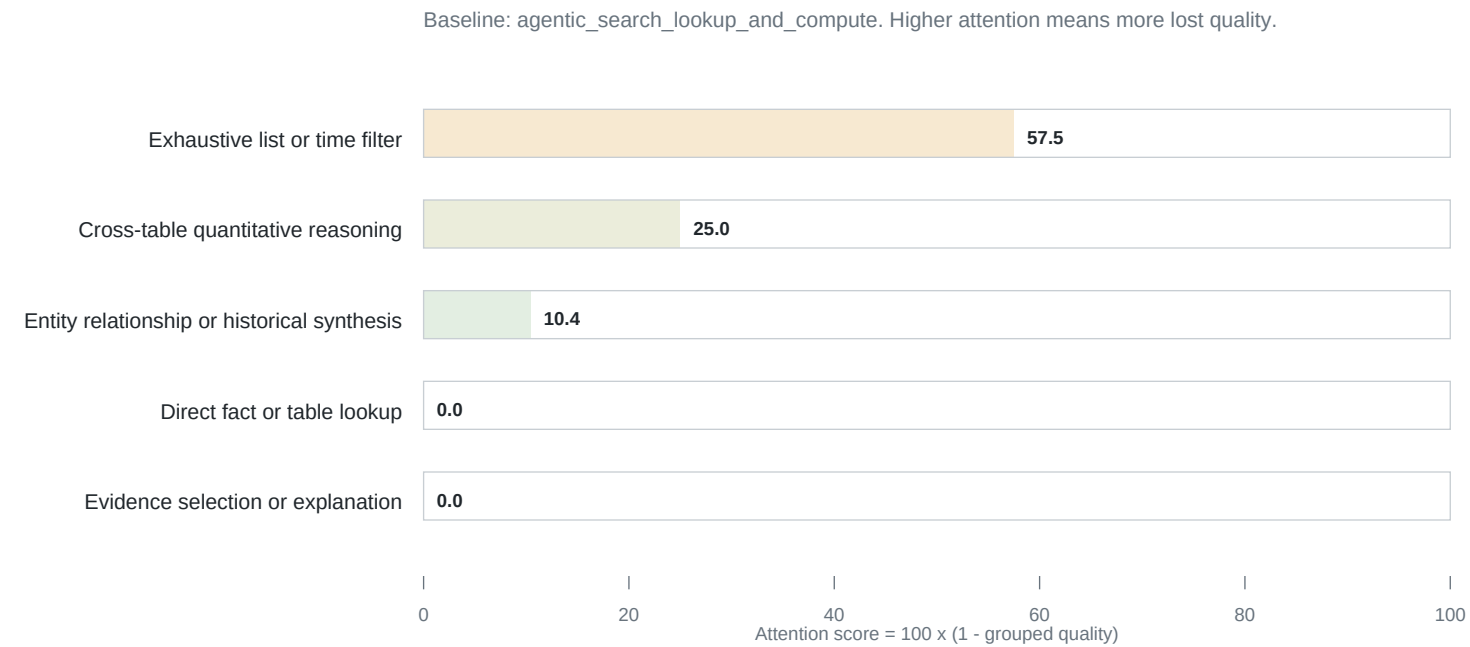
Fraction of each question's designated source articles that appeared in the retrieved set (retriever configurations only). 1.00 = every gold article retrieved; lower values mark a retrieval miss — usually a multi-source question.

Question	agentic_retriever_with_toc	agentic_search_lookup_and_compa	agentic_toc_tables_as_is	agentic_toc_with_table_computation	s_k7_md_graph_n5retriever
GT01	100.0%	100.0%	100.0%	100.0%	100.0%

Question	agentic_retriever_with_toc	agentic_search_lookup_and_compa	agentic_toc_tables_as_is	agentic_toc_with_table_computation	s_k7_md_graph_n5retriever
GT02	100.0%	100.0%	100.0%	100.0%	100.0%
GT03	100.0%	100.0%	100.0%	100.0%	33.3%
GT04	100.0%	100.0%	100.0%	100.0%	100.0%
GT05	100.0%	100.0%	100.0%	100.0%	100.0%
GT06	100.0%	100.0%	100.0%	100.0%	100.0%
GT07	100.0%	100.0%	100.0%	100.0%	100.0%
GT08	100.0%	100.0%	100.0%	100.0%	100.0%
GT09	100.0%	100.0%	100.0%	100.0%	50.0%
GT10	100.0%	100.0%	100.0%	100.0%	100.0%
GT11	100.0%	100.0%	100.0%	100.0%	100.0%
GT12	100.0%	100.0%	100.0%	100.0%	100.0%
GT13	100.0%	100.0%	100.0%	100.0%	100.0%
GT14	100.0%	100.0%	100.0%	100.0%	100.0%
GT15	100.0%	100.0%	100.0%	100.0%	100.0%
GT16	100.0%	100.0%	100.0%	100.0%	100.0%
GT17	100.0%	100.0%	100.0%	100.0%	100.0%
GT18	100.0%	100.0%	100.0%	100.0%	100.0%
GT19	50.0%	100.0%	50.0%	100.0%	100.0%

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.



Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.425	57.5	55.0%	n/a	n/a	100.0%
Cross-table quantitative reasoning	2	0.750	25.0	0.0%	n/a	n/a	100.0%
Entity relationship or historical synthesis	6	0.896	10.4	8.3%	n/a	n/a	100.0%
Direct fact or table lookup	3	1.000	0.0	0.0%	n/a	n/a	100.0%
Evidence selection or explanation	3	1.000	0.0	0.0%	n/a	n/a	100.0%

Baseline quality and failure rate use agentic_search_lookup_and_compute. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

	agentic_retriever_with_toc	agentic_search_lookup_and_compute	agentic_toc_tables_as_is	agentic_toc_with_table_computation	vs_baseline_md_graph_n5retriever
1 article	**0.591**	**0.830**	**0.682**	**0.682**	**0.727**
2 articles	**0.575**	**0.775**	**0.725**	**0.750**	**0.867**
3+ articles	**0.542**	**0.667**	**0.708**	**0.667**	**0.667**

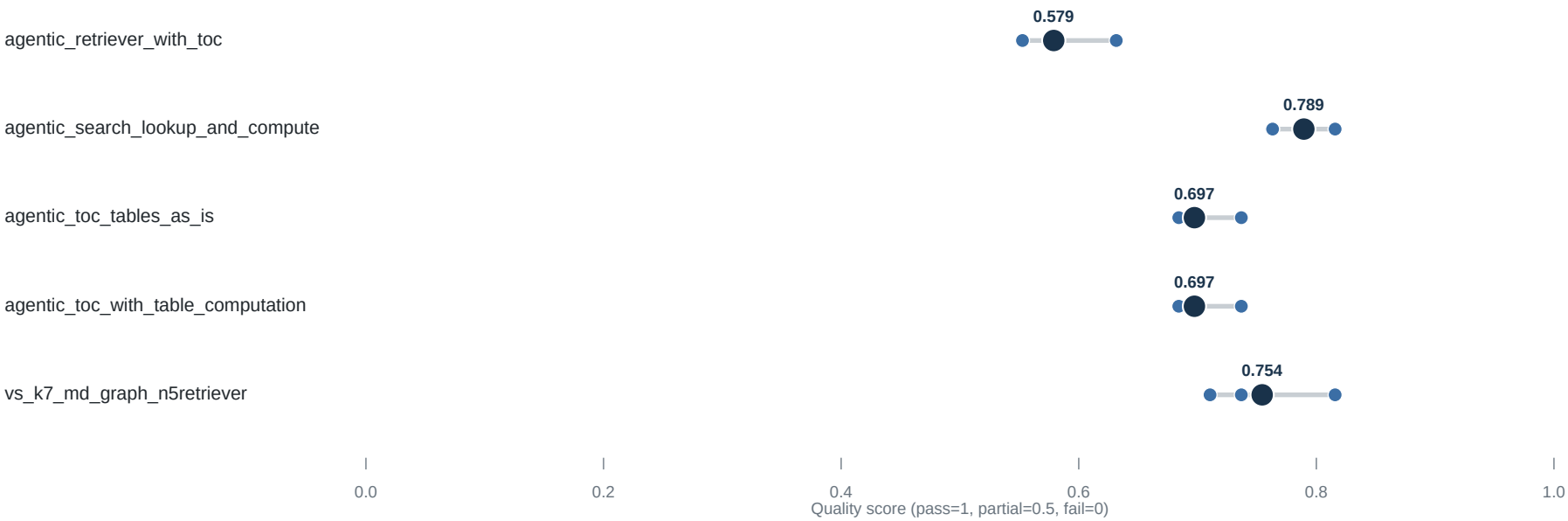
Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	agentic_retriever_with_toc	11	44	47.7%	22.7%	29.5%	0.591	100.0%
2 articles	agentic_retriever_with_toc	5	20	25.0%	65.0%	10.0%	0.575	90.0%
3+ articles	agentic_retriever_with_toc	3	12	41.7%	25.0%	33.3%	0.542	100.0%
1 article	agentic_search_lookup_and_compute	11	44	81.8%	2.3%	15.9%	0.830	100.0%
2 articles	agentic_search_lookup_and_compute	5	20	65.0%	25.0%	10.0%	0.775	100.0%
3+ articles	agentic_search_lookup_and_compute	3	12	66.7%	0.0%	33.3%	0.667	100.0%
1 article	agentic_toc_tables_as_is	11	44	59.1%	18.2%	22.7%	0.682	100.0%
2 articles	agentic_toc_tables_as_is	5	20	50.0%	45.0%	5.0%	0.725	90.0%
3+ articles	agentic_toc_tables_as_is	3	12	66.7%	8.3%	25.0%	0.708	100.0%
1 article	agentic_toc_with_table_computation	11	44	63.6%	9.1%	27.3%	0.682	100.0%
2 articles	agentic_toc_with_table_computation	5	20	55.0%	40.0%	5.0%	0.750	100.0%
3+ articles	agentic_toc_with_table_computation	3	12	66.7%	0.0%	33.3%	0.667	100.0%

Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	vs_k7_md_graph_n5retriever	11	33	69.7%	6.1%	24.2%	0.727	100.0%
2 articles	vs_k7_md_graph_n5retriever	5	15	80.0%	13.3%	6.7%	0.867	90.0%
3+ articles	vs_k7_md_graph_n5retriever	3	9	66.7%	0.0%	33.3%	0.667	77.8%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
eloquent_bezos	agentic_retriever_with_toc	19	9	3	7	0.553	97.4%	94.7%
trusting_turing	agentic_retriever_with_toc	19	7	7	5	0.553	97.4%	94.7%
zen_cerf	agentic_retriever_with_toc	19	7	8	4	0.579	97.4%	94.7%
naughty_carmack	agentic_retriever_with_toc	19	8	8	3	0.632	97.4%	94.7%
gifted_goldwasser	agentic_search_lookup_and_compute	19	14	3	2	0.816	100.0%	100.0%
zealous_stallman	agentic_search_lookup_and_compute	19	14	1	4	0.763	100.0%	100.0%
upbeat_page	agentic_search_lookup_and_compute	19	14	1	4	0.763	100.0%	100.0%
boring_jobs	agentic_search_lookup_and_compute	19	15	1	3	0.816	100.0%	100.0%
sweet_hopper	agentic_toc_tables_as_is	19	10	6	3	0.684	97.4%	94.7%

Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
eloquent_lovelace	agentic_toc_tables_as_is	19	11	4	4	0.684	97.4%	94.7%
brave_bezos	agentic_toc_tables_as_is	19	11	4	4	0.684	97.4%	94.7%
hardcore_wing	agentic_toc_tables_as_is	19	12	4	3	0.737	97.4%	94.7%
awesome_cook	agentic_toc_with_table_computation	19	12	2	5	0.684	100.0%	100.0%
blissful_lamport	agentic_toc_with_table_computation	19	11	4	4	0.684	100.0%	100.0%
pedantic_tanenbaum	agentic_toc_with_table_computation	19	12	4	3	0.737	100.0%	100.0%
cool_ritchie	agentic_toc_with_table_computation	19	12	2	5	0.684	100.0%	100.0%
bold_wozniak	vs_k7_md_graph_n5retriever	19	13	2	4	0.737	93.9%	89.5%
awesome_valiant	vs_k7_md_graph_n5retriever	19	13	1	5	0.711	93.9%	89.5%
reverent_wilkes	vs_k7_md_graph_n5retriever	19	15	1	3	0.816	93.9%	89.5%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by a CSV experiment_name or config_name when one is present. If no explicit name is present, grouping falls back to evaluation mode, context source, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
awesome_cook.csv
awesome_valiant.csv
blissful_lamport.csv
bold_wozniak.csv
boring_jobs.csv
brave_bezos.csv
cool_ritchie.csv
eloquent_bezos.csv
eloquent_lovelace.csv
gifted_goldwasser.csv
hardcore_wing.csv

Run file
naughty_carmack.csv
pedantic_tanenbaum.csv
reverent_wilkes.csv
sweet_hopper.csv
trusting_turing.csv
upbeat_page.csv
zealous_stallman.csv
zen_cerf.csv